

PRANAY DATTA KAVUKUNTLA

DATA ENGINEER

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SUMMARY

- Experienced Data Engineer with over 3+ years of expertise in designing, building, and optimizing complex data pipelines, ETL processes, and scalable data architectures tailored for large-scale data environments and high-performance analytics.
- Proficient in big data frameworks like Hadoop and Apache Spark for efficient distributed data processing, along with expertise in SQL, Python, and SSIS for seamless data integration, transformation, and in-depth analysis.
- Leveraged cloud platforms like Microsoft Azure (Data Factory, SQL Database, Synapse Analytics), AWS (S3, EC2, Lambda), and GCP (BigQuery, Dataflow) to implement scalable solutions for efficient real-time and batch data processing across diverse environments.
- Skilled in building automated data pipelines and implementing CI/CD pipelines for ETL workflows, effectively reducing manual intervention, improving overall efficiency, and significantly enhancing data reliability across diverse environments and use cases.
- Demonstrated ability in data modeling, data warehousing, and schema design, ensuring optimal data organization and accessibility for analytics and business intelligence, ultimately facilitating informed decision-making and strategic insights.
- Expert in data quality assurance and governance, using tools like Azure Data Factory, Apache NiFi, and Talend to monitor, validate, and cleanse data, ensuring consistency and accuracy across multiple sources while enhancing overall data integrity and reliability.
- Proficient in developing interactive and insightful data visualizations using Power BI, Tableau, and Qlik Sense, enabling stakeholders to effectively interpret complex datasets and facilitating data-driven decision-making across organizations.
- Successfully led data migration and integration initiatives across diverse platforms, ensuring optimized performance, minimal downtime, and seamless transitions, particularly in high-volume environments, leading to more efficient data management.
- Skilled at handling structured and unstructured datasets, employing SQL, Python scripting, and automation on platforms like Snowflake, Azure, and GCP to clean, extract, and transform data from diverse sources, enhancing data accessibility and quality.
- Proven track record in improving data infrastructure, streamlining ETL processes, and reducing operational costs through effective automation strategies and cloud-based solutions across Azure, AWS, GCP, and Snowflake, driving operational efficiency.

TECHNICAL SKILLS

Programming Skills:	SQL, PL/SQL, Python
Cloud Technologies:	Microsoft Azure (Azure Data Factory, Azure SQL Database, Azure Synapse Analytics), Google Cloud Platform (Big Query, Dataflow), AWS, Snowflake
ETL Tools:	SSIS, Azure Data Factory.
Data Visualization Tools:	Tableau, Qlik Sense, Microsoft Power BI, MS Excel (Advanced)
Database:	MySQL, SQL Server, PostgreSQL, NoSQL, MongoDB
Big Data & Analytics:	Apache Spark, Hadoop, HDFS, Hive, Apache Kafka, Apache Atlas, Sqoop, Databricks.
Database Management:	SQL Server, MySQL, HDFS.
Data Monitoring & Automation:	Azure Monitor, Apache Atlas, SQL Server Agent.
Libraries & Version Control	Pandas, Matplotlib, NumPy, Seaborn, Sci-kit learn, TensorFlow, Git, GitHub

PROFESSIONAL EXPERIENCE

Marlabs Inc (Elsevier), CA

Data Engineer Intern

Jul 2024 – Current

- Developed and optimized data pipelines with Azure Data Factory and Databricks, streamlining workflows for large-scale analytics.
- Integrated data sources using Apache Kafka and Azure Event Hubs, enabling real-time ingestion and processing for critical operations.
- Improved ETL processes with SQL and Python by automating data validation and error handling, ensuring high data integrity.
- Designed and deployed custom monitoring dashboards in Power BI to track pipeline performance and error rates in real-time, facilitating proactive issue resolution and significantly enhancing overall data pipeline reliability and operational efficiency.
- Implemented automated data schema detection and adjustment features, significantly reducing manual intervention and ensuring seamless data transformation across various datasets, which led to improved workflow efficiency and enhanced data integrity.
- Developed data partitioning and indexing strategies in Azure Synapse Analytics, optimizing query performance and retrieval speed.
- Built data lineage systems with Apache Atlas and Azure Purview to enhance transparency and ensure compliance with data standards.
- Implemented Snowflake solutions to optimize storage and query performance, ensuring integration with data and boosting analytics.
- Leveraged Snowflake’s data sharing features for real-time access across teams, enhancing data-driven decision-making.
- Integrated CRM data in Palantir Foundry, creating unified customer profiles for enhanced accessibility and large-scale analytics.
- Built automated pipelines in Salesforce for real-time customer insights, greatly enhancing team decision-making accuracy.

LTIMindtree (Sony & Ralph Lauren), Hyderabad, India

Data Engineer

Oct 2020 – Jul 2022

- Engineered and maintained SSIS packages for SQL Server data integration with legacy systems, enabling seamless data consolidation and creating efficient workflows for high-profile clients like Sony and Ralph Lauren, ensuring smooth data operations.
- Optimized SQL stored procedures and indexes within SSMS, leading to significant system performance improvements, reducing query execution times, and boosting overall efficiency by 27%, facilitating quicker data access and processing.
- Executed a cloud migration strategy using SSIS and SSMS, implementing automated data transfer protocols and efficient cloud database management solutions, resulting in cost savings and enhanced data accessibility across cloud environments.

- Led comprehensive data quality improvement initiatives using SSIS, incorporating advanced data cleansing, validation, and transformation techniques to maintain high levels of data accuracy and consistency across all integrated systems, minimizing errors.
- Employed Azure Data Factory to automate ETL workflows for SQL Server data integration, significantly reducing manual effort, ensuring seamless data consolidation, and enabling consistent data processing for clients like Sony and Ralph Lauren.
- Migrated and transformed financial data from SQL Server to HDFS using Sqoop, facilitating scalable storage solutions and optimizing data access for more efficient downstream analytics, reporting, and processing.
- Designed and implemented a distributed data processing framework using Apache Spark and Python, enabling parallel processing and significantly accelerating data ingestion and transformation by 25%.
- Developed advanced data visualizations with Plotly and Seaborn in Python, translating complex datasets into actionable insights.
- Integrated a real-time data monitoring system in Power BI, enabling stakeholders to track KPIs and make data-driven decisions.
- Built a robust metadata management system using Apache Atlas, ensuring data governance and compliance with industry standards.
- Implemented incremental data loading in SSIS, minimizing load times and reducing resource consumption during ETL operations.
- Leveraged both Azure Synapse Analytics and Snowflake to handle diverse data workloads, utilizing Azure for data storage and transformation, while employing Snowflake for advanced analytics and scalable querying.
- Collaborated with stakeholders to define data requirements and ensure data quality for optimized integration and analytics.

Adani, India
Data Engineer

Jan 2020 – Sep 2020

- Designed interactive visualizations with Matplotlib and Plotly, enabling stakeholders to gain insights and improve decision-making.
- Employed MongoDB for unstructured data storage and analysis, expanding capabilities and driving insights from diverse sources.
- Created interactive visualizations in Power BI, improving data-driven storytelling and supporting better stakeholder communication.
- Collaborated with cross-functional teams using Agile to ensure timely delivery of data projects and improve processes.
- Optimized data workflows by integrating data pipelines with AWS services such as S3 and Lambda, significantly reducing processing time and enhancing overall data reliability, leading to smoother data operations and faster insights.
- Developed robust ETL pipelines using Python and SQL to automate the processing of financial data, reducing integration time by 30% and significantly improving data accuracy across multiple systems, resulting in more reliable and efficient data management.

Data Analyst Intern

Jul 2019 – Dec 2019

- Developed Python scripts to automate data processing tasks, improving workflow efficiency and data manipulation capabilities.
- Executed complex SQL queries to extract, analyze, and interpret data, supporting business intelligence and improving data accuracy.
- Developed Tableau dashboards and reports to highlight key performance indicators, aiding improved strategic planning and decisions.
- Conducted data cleansing and preprocessing using MS Excel, ensuring the accuracy and integrity of datasets for thorough analysis.

PROJECTS

Parkinsons Disease detection using Machine Learning

- Collected and preprocessed vocal datasets from Parkinson's patients and controls, extracting jitter, shimmer, and pitch features.
- Built SSIS packages and optimized workflows with Apache Spark and SQL functions for efficient processing and high-quality inputs.
- Trained XGBoost models on preprocessed data, achieving high sensitivity and specificity in Parkinson's disease detection.
- Utilized Apache Kafka for real-time data streaming and processed vocal data, improving data consistency and quality.
- Created an online interface for vocal data input and analysis, deploying models with Docker and managing version control with Git.
- Visualized data with Matplotlib and collaborated with cross-functional teams using Jira and Slack to ensure project success.

Crime Data Analysis in Worcester

- Led Tableau crime data visualizations, using plots and graphs to highlight trends and enhance public safety strategies in Worcester.
- Conducted time-series analysis in Tableau, visualizing crime hotspots and trends for optimized safety initiatives.
- Utilized GIS tools like QGIS and ArcGIS for geospatial visualizations, enhancing insights into crime data.
- Employed Python libraries such as Pandas, NumPy, and Matplotlib for in-depth data analysis and creating custom visualizations.
- Streamlined data visualization processes by integrating data from Azure into interactive dashboards for real-time analysis.

Advanced Data Visualization for Vaccine Safety Using CDC VAERS

- Analyzed and transformed CDC's VAERS dataset to identify trends in vaccine-related adverse events across demographics.
- Preprocessed and cleaned datasets by handling missing data, normalizing values, and ensuring consistency for accurate analysis.
- Developed Tableau and QlikView dashboards for real-time, audience-specific data exploration and insights.
- Created visualizations like risk analysis bar charts to highlight key trends in vaccine safety and event frequency.
- Applied advanced filtering techniques to enable dynamic, interactive data exploration by age, gender, and vaccine type.

Data Pipeline Optimization for Market Analysis and Segmentation

- Designed and implemented ETL processes with SSIS to extract, transform, and load data from multiple sources, boosting efficiency.
- Applied Python and SQL for data cleansing, validation, and enrichment, improving data quality and reducing inconsistencies by 25%.
- Created interactive dashboards in QlikView and Python to analyze market trends and customer behavior, improving decision-making.
- Performed segmentation analysis and calculated acquisition metrics, leading to improved acquisition rates and reduced costs.
- Integrated dynamic filters, parameters, and drill-downs into dashboards, enhancing data exploration and accessibility for analysis.

EDUCATION

M.S. Computer Science, Clark University, Worcester, MA	Aug 2022 – May 2024
B.E Information Technology, Chaitanya Bharati Institute of Technology, Hyderabad, India	Aug 2016 – Apr 2020